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Read ppts compulsory of final term for very best performance in the final exams it will be very good for your excellence in bt404 final term exams in_SHAA-ALLAH

What are predominant Microorganisms

Unspoiled, nonsterile food generally contains many types of microorganisms from different
genera.
When the same food is spoiled, it is found to contain predominantly one or two types and they
may not even be present initially in the highest numbers in the unspoiled fresh product.

1. **relationships between Microorganisms and food.**
   elationship of Microorganisms and food
   • **Pathogenic or Toxogenic: cause disease**
   • **Beneficial/ desirable**(make food)
   – Lactobacilli bacteria in fermented food(yoghurt, pickles)
   – Yeast Saccharomyces in Bread and baking
   • **Deteriorated/ undesirable**
   – Non-pathogenic: Spoilage
   – Fungi, Pseudomonas, Proteus, Bacillus
   Conditions for Spoilage
   • Water, pH ,physical structure, oxygen, temperature
   3. What is the role of Microorganisms.

Beneficial **Role of Microorganisms in Food** Industry. **Microorganisms** play an important **role** in **food** industry. ... The **microorganisms** that cause **food** spoilage

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and also find the maximum exploitation in production of **food** and **food** products are mainly **bacteria**, molds and yeasts. Transcript of The **Role of Microorganisms in Food Production**. **microorganisms**to carry out fermentation processes, and for thousands of years mankind has used yeasts, moulds and **bacteria** to make **food** products such as bread, beer, wine, vinegar, yoghurt and cheese, as well as fermented fish, meat and vegetables

What is food safety. 5 marks

What is food safety?
• Potential health risks from food consumption
– microbiological, viral and parasitic concerns
– hormone residues (growth promoters)

animal drugs (antibiotics)
– chemical residues (pesticides)

preservatives

bioengineered foods
• Americans have the safest food supply in the world
• Food industry highly regulated
• Inspections at:
– Antemortem (before death)
– Postmortem (after death)
– Further processing
– Facilities
– Stores
– Restaurant kitchens
– Americans have the safest food supply in the world
• Inspections performed by:
– Federal agencies
– State agencies
– Local agencies
– Meat Industry Examples

• Federal Agencies
• Meat Industry Examples

Federal Agencies
– Food Safety and Inspection Service (FSIS)
• Meat Industry Examples
• Federal Agencies
– Food Safety and Inspection Service (FSIS)
– Animal, Plant, and Health Inspection Service (APHIS
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– Grain Inspection, Packers and Stockyard Administration (GIPSA)
• Meat Industry Examples
• Federal Agencies
– Food Safety and Inspection Service (FSIS)
– Animal, Plant, and Health Inspection Service (APHIS)
– Grain Inspection, Packers and Stockyard Administration (GIPSA)
– Agricultural Marketing Service (AMS)

Write the names of food safety programs and explain each point. 10 marks.

A **food safety program** is a written document indicating how a **food** business will control the **food safety** hazards associated with the **food** handling activities of the business

Food borne infection and toxification.

Food Safety Programs
• HACCP
H: Hazard
A: Analysis
C: Critical
C: Control
P: Point

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HACCP Plans
• Assemble HACCP team
• Describe the food and the method of distribution
• Identify the intended use and consumers of the food
• Develop a flow diagram which describes the process
• Verify the flow diagram
• HACCP Program Principles
1 Conduct a hazard analysis
• Biological hazard: Ex. Bacteria
• Chemical hazard: Ex. Nitrite toxicity
• Physical hazard: Ex. Metal

HACCP Program Principles
1 Conduct a hazard analysis
2 Identify the Critical Control Points (CCP) in the process
Ex. Endpoint temperature
• HACCP Program Principles
1 Conduct a hazard analysis
2 Identify the Critical Control Points (CCP) in the process
Ex. Endpoint temperature
3 Establish limits for preventative measures associated with each CCP

Ex. Reach temperature of 140oF
• HACCP Program Principles
3 Establish limits for preventative measures associated with each CCP
4 Establish CCP monitoring requirements EX. thermometer
5 Establish corrective action to be taken when monitoring indicates a critical limit deviation
• HACCP Program Principles
6 Establish effective record keeping procedures that document the HACCP system
7 Establish procedures for verification that the HACCP system is working
• Pathogenic Bacteria Concerns
• Food poisoning:
• Food infection:

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• Pathogenic Bacteria Concerns
• Food poisoning: illness caused by the ingestion of toxins
• Food infection: ingestion of pathogenic organisms that grow and cause illness
• Microbiological Concerns
• ***Clostridium botulinum***: food poisoning; botulism; low acid foods, anaerobic, sodium
nitrite helps control
• Microbiological Concerns
• ***Clostridium botulinum***: food poisoning, botulism, low acid foods, anaerobic, sodium
nitrite helps control
• ***Staphylococcus aureus***: food poisoning, faculative, food handling problem, proper
temperature helps control
• Microbiological Concerns
• ***Salmonella***: carried intestinal tracts, improper cooking, avoid cross contamination
• Microbiological Concerns
• ***Salmonella***: carried intestinal tracts, improper cooking, avoid cross contamination
• ***Clostridium perfringens***: anaerobic, produces an enterotoxin, 'cafeteria germ', keep food
>140oF
• Microbiological Concerns

Food borne infection:

Food borne infection is the infection which is caused by food borne bacteria . Following are the types of food borne infections.

Two primary types
– **Food infection** : due to presence of living organism
– **Food poisoning** due to presence of toxins

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Infections
• Bacterial / Viral / parasite
• Invade and or multiply in lining of intestine
• Incubation period- hours to days
• S/s – Diarrhea , nausea, vomiting , abdominal cramps, fever
• Communicable-spreads from person to person
• Factors-inadequate cooking, cross contamination , poor personal hygiene , bare hand
contact

What is steam pasteurization?

Steam Pasteurization: whole carcass treatment; brings surface to about 200oF; extremely effective; expensive and water and energy intensive

What is food additives?

Food additives are substances added to products to perform specific technological functions. These functions include preserving, i.e. increasing shelf-life or inhibiting the
growth of pathogens, or adding colouring and flavouring to food for interest and variety.
• There are over 300 permitted additives that can be used in the UK. Flavourings are not
included in this figure, as there are over 3,000 flavouring components in UK use, in many
different combinations. International organisations provide advice on the safety of flavourings.

Centers of food and applied nutrition of following kind? Kinds are not mentioned .

Center for Food Safety and Applied Nutrition
■□Center for Veterinary Medicine
■□Feed Safety

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■ Animal Health/Protection
■ Office of Regulatory Affairs
■ Detection method adaptation
■ Methods validation
■ FERN
■ National Center for Toxicological Research
■ "Foods" Research Program
■ Center for Food Safety and Applied Nutrition
■ Methods development
■ Agent characterization
• Pathogencity/toxicity
• Behavior in foods
■ Intervention technologies
■ Toxicity/Pathogenicity
■ Attempted to enhance food safety while advancing food security
■ "Foods" Research Program
■ Leverage research performed by medical, agricultural, and defense research agencies and
the academic community
■ DHHS NIH and CDC
■ USDA ARS and CREES
■ DOD
■ CFSAN Intramural Research Program
■ Meeting Critical Laboratory Infrastructure Needs
■ Training of research to work with non-traditional agents
■ Upgrading of laboratory security

Name the food safety research programs?
Difference between infection and intoxication?

Infections
• Bacterial / Viral / parasite
• Invade and or multiply in lining of intestine
• Incubation period- hours to days

• S/s – Diarrhea , nausea, vomiting , abdominal cramps, fever
• Communicable-spreads from person to person
• Factors-inadequate cooking, cross contamination , poor personal hygiene , bare hand contact intoxications
• toxins (natural / preformed bacterial / chemical)
• No invasion or multiplication
• Incubation period- minutes to hours
• S/s – Vomiting , nausea, diarrhea , diplopia, weakness, resp. failure , numbness, sensory/motor dysfunction

What is food safety?

Food safety refers to the conditions and practices that preserve the quality of **food**to prevent contamination and **food**-borne illnesses. The **Food Safety** and Inspection Service of the USDA educates consumers about the importance of **safe food**handling and how to reduce the risks associated with foodborne illness.

Write the names of food safety programs? Explain each points.
Commodity Cobditions for food biotechnology detection methods

Address priority agent/commodity combinations
■ Detection methodology
■ Agent characteristics
■ Prevention/intervention technologies
■ Rapid response and recovery capabilities
■ Seek additional resources to accelerate progress
■ The Future
■ Keeping focused on the goal: Completing as rapidly as possible the critical translational
research that FDA needs to assure the security of the nation's food supply

Role of Radiation in Food safety

Irradiation:
– "Electronically Pasteurized"
– Costly
– Consumer resistance

– 3 types:
• Gamma Ray
• Electron-Beam
• X-Ray
– What is the industry doing to control pathogens?
• Intervention process are not a substitute for Good Manufacturing Practices (GMP) and
Sanitation Standard Operating Procedures (SSOP)

How to prevent food Borne illness

Prevention:because of the seriousness of the illness, the accompanying avoidance measures have been actualized at the gathering point, which has dispensed with any conceivable transmission of the malady to general society. All conceivable contaminated parts of these creatures are evacuated at the reaping plant, kept separate from all other creature waste, and obliterated to shield the nourishment supply. The parts evacuated incorporate the skull, tonsils, a thick cut of the focal spine including the spinal segment from the base of the skull to the pelvis, and two segments of the small digestion tracts.

Center of food safety and applied nutrition have center?
Relationship of microorganisms and food
Food Borne Intoxication and infection
Detection methodology in primary food research with the result, food microbiology laboratories now have a wide array of detection methods and automated technologies such as enzyme immunoassay, polymerase chain reaction, and microarrays, which can cut test times considerably. Nucleic acid amplification strategies and advances in amplicon detection methodologies have been the key factors in the progress of molecular microbiology. A comprehensive literature survey has been carried out to give an overview in the field of foodborne pathogen detection. In this paper, we describe the conventional methods, as well as recent developments in food pathogen detection, identification, and quantification, with a major emphasis on molecular detection methods.
How Global food safety works

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The Global Food Safety Initiative (GFSI) was set up in 2000 as a sub-group of the Consumer Goods Forum.

About GFSI: *What is it?*

• GFSI provides a platform for collaboration between some of the world's leading
food safety experts from retailer, manufacturer and food service companies, service
providers associated with the food supply chain, international organizations, academia and government.

• GFSI developed a model that determines equivalency between existing food safety
schemes, whilst leaving flexibility and choice in the marketplace.

• Objectives:

• Reduce food safety risks by delivering equivalence and **convergence** between effective
food safety management systems.

• Manage cost in the global food system by **eliminating redundancy** and improving
operational efficiency.

• Develop competencies and capacity building in food safety to create consistent and
effective global food systems.

• Provide a unique international stakeholder platform for collaboration, knowledge exchange and networking.

• GFSI remains focused on food safety.

• GFSI does not deal with environmental or social issues

Name the food safety programs and write two points